

Week 14-15 Office Hour Notes

14.0 Errata

14.1 Completeness Theorem Overview

Now, we shall prove the converse of soundness, which is known as the **Completeness Theorem**:

Theorem 14.1.1. (*Completeness Theorem part I*) The converse of soundness holds, i.e., $\Gamma \models \varphi \implies \Gamma \vdash \varphi$.

Theorem 14.1.2. (*Completeness Theorem part II*) If Γ is consistent, then Γ is satisfiable and a model $\mathfrak{A}$ of Γ exists.

Lemma 14.1.3. **Theorem 14.1.1** $\iff$ **Theorem 14.1.2.**

Proof. Left as an exercise for the homework of Week 14-15. $\square$

%MAKE STATEMENT COMPARING LOGICAL IMPLICATION AND TAUTOLOGICAL IMPLICATION TO OPEN UP WEEK 16 OFFICE HOURS

Recall that satisfiability in first order logic is when there exists a model $\mathfrak{A}$ of Γ , i.e. structure $\mathfrak{A}$ and a variable assignment function $s : V \rightarrow |\mathfrak{A}|$ such that $\models_{\mathfrak{A}} \gamma[s]$, for all $\gamma \in \Gamma$ (which is different from satisfiability in the sense of there existing a variable assignment function $v : \{A\}_{n \in \mathbb{N}} \rightarrow \{T, F\}$)

We define finite satisfiability analogously as follows:

Definition 14.1.4. Γ is said to be **finitely satisfiable** if for every finite $\Gamma_0 \subset \Gamma$, there exists a structure $\mathfrak{A}_0$ and a variable assignment function $s : V \rightarrow |\mathfrak{A}_0|$ such that $\models_{\mathfrak{A}_0} \gamma[s]$, for all $\gamma \in \Gamma_0$ (and not necessarily all of Γ)

In other words, every finite $\Gamma_0 \subset \Gamma$ has a model $\mathfrak{A}_0$.

Corollary 14.1.5. (*Compactness Theorem for First order Logic*) Γ is satisfiable if and only if Γ is finitely satisfiable.

Proof. $\implies$ Immediate since we can choose a model $\mathfrak{A}$ of Γ and $\mathfrak{A}$ by definition satisfies

every finite $\Gamma_0 \subset \Gamma$.

$\Leftarrow$ Suppose Γ is finitely satisfiable. By completeness (**Theorem 14.1.2**), it suffices to show that Γ is satisfiable by showing that Γ is consistent. To do this, suppose towards contradiction that Γ is inconsistent. Then there exists β such that $\Gamma \vdash \beta$ and $\Gamma \vdash \neg\beta$. Choose deductions S_1, S_2 that prove β and $\neg\beta$, respectively, set

$$\Gamma_0 := \left\{ \gamma \in \Gamma : \gamma \text{ is utilized either of the deductions } S_1 \text{ or } S_2 \right\},$$

and note that since Γ_0 is a subset of all formulas used in the deductions of S_1 and S_2 (which are finite sequences), we find Γ_0 is finite, and we have $\Gamma_0 \vdash \beta$ and $\Gamma_0 \vdash \neg\beta$, since we can utilize the deduction S_1 and S_2 , respectively. It follows by soundness (**Theorem 13.2.2**) that $\Gamma_0 \models \beta \wedge \neg\beta$. Since $\beta \wedge \neg\beta$ is a contradiction, Γ_0 is unsatisfiable, however, this contradicts finite satisfiability, completing the proof. $\square$

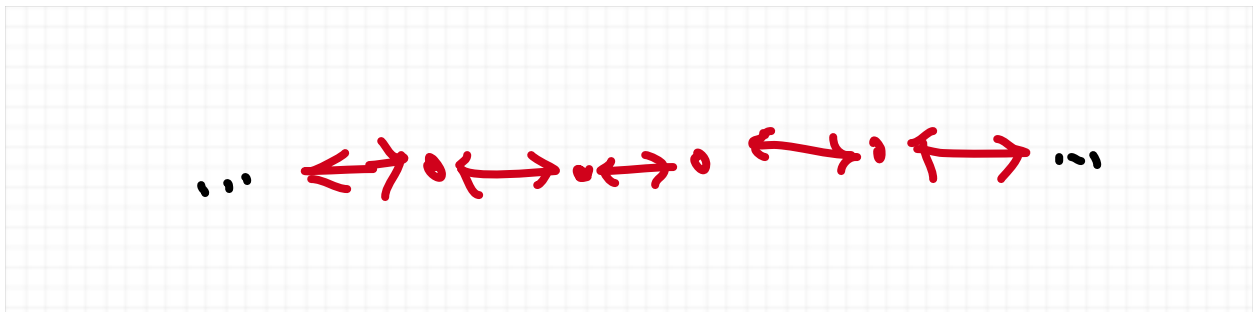
15.0 Errata

15.1 Questions

%PROVIDE HINT FOR PROBLEM 2 OF HOMEWORK 13

Question 1. A lot students had questions on **Question 3 of Homework 13** due this Monday.

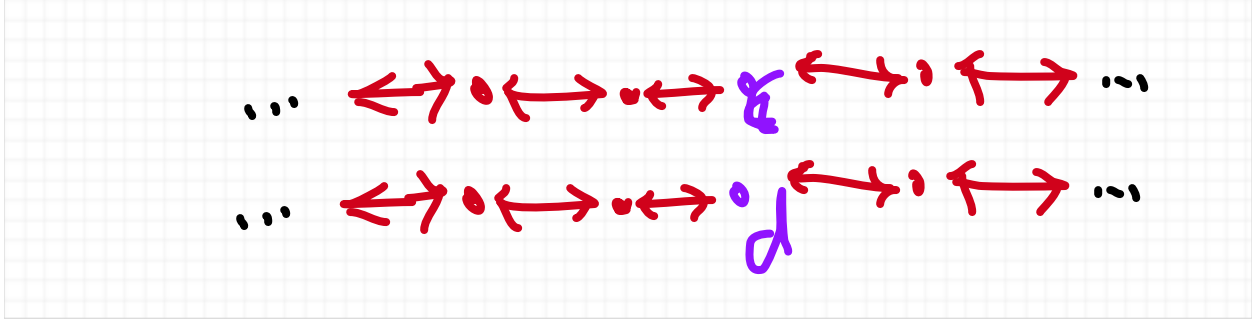
Recall that the question asks that in the language of graphs (explained in the problem), if we have a structure $\mathfrak{U}$ with $|\mathfrak{U}| = \mathbb{Z}$, and the graph constructed as follows



We want to show that there exists a disconnected structure $\mathfrak{B}$ elementarily equivalent to $\mathfrak{A}$.

Intuitive Idea: The language we use is able to talk about paths of finitely large lengths not

existing, but the language isn't able to talk about the paths existing or not existing in a single formula. That begs the question as to whether a graph G that consists of two disjoint copies of $\mathfrak{A}$ suffices as an elementarily equivalent structure.



But the question is how do we go about showing that they're elementarily equivalent?

The tool we have at our disposal is compactness, but in order to use compactness, we need do the following step-by-step process:

Step 1. Extend the language $\mathcal{L} = \{P\}$ in the original problem to $\mathcal{L}' := \mathcal{L} \cup \{c, d\} = \{P, c, d\}$, where c and d are constant symbols.

Step 2. Construct a scheme of formulas in $\mathcal{L}'$ that when all true establish that c and d are disconnected, which is done in the hint with ϕ_n , for each $n \geq 1$ constructed as so:

$$\phi_n := \neg \exists x_1 \cdots \exists x_{n-1} (P(c, x_1) \wedge P(x_1, x_2) \wedge \cdots \wedge P(x_{n-2}, x_{n-1}) \wedge P(x_{n-1}, d))$$

Step 3. Show that the set

$$\Gamma := \text{Th}_{\mathcal{L}}(\mathfrak{A}) \cup \{\phi_n : n \geq 1\},$$

where

$$\text{Th}_{\mathcal{L}}(\mathfrak{A}) := \left\{ \text{well-formed } \mathcal{L}\text{-sentences } \varphi \text{ (note that they're NOT } \mathcal{L}'\text{-sentences)} : \models_{\mathfrak{A}} \varphi \right\}$$

is finitely satisfiable. This is the step that will be left to the reader, and it involves taking an arbitrary finite $\Gamma_0 \subset \Gamma$ and constructing a model of Γ_0 .

Step 4. Use the **Compactness Theorem (Corollary 14.1.5)** to show that there exists a model $\mathfrak{B}$ of Γ .

Step 5. Show elementary equivalence (in the language $\mathcal{L}$) between the two structures $\mathfrak{A}, \mathfrak{B}$ recalling that two structures are elementary equivalent (writing $\mathfrak{A} \equiv \mathfrak{B}$) if for every sentence σ , we have

$$\models_{\mathfrak{A}} \sigma \iff \models_{\mathfrak{B}} \sigma.$$

Note that details of **Step 3-5** are left to the reader.

15.2 Completeness Theorem Proof

Important Note: In this proof, we assume that the original language $\mathcal{L}$ is countable, since while we can prove the more general completeness theorem when $\mathcal{L}$ is uncountable, we then need to use somewhat advanced Set Theory that is in general beyond the scope of the course, but I'll give an overview of it in the next Optional Office Hour this Monday 3pm-4:30pm.

General Outline of Proof of the Completeness Theorem (Theorem 14.1.1). We shall do it using the following steps:

Step 0. To use equivalence (**Lemma 14.1.3**) of **parts 1 and 2** of the **Completeness Theorem** as stated in the previous section (**Theorem 14.1.1** and **Theorem 14.1.2**) to state that it suffices to show that if Γ is consistent, then Γ is satisfiable.

Next, we use **Steps 1-3** (as we state later) to prove that there exists an expanded language $\mathcal{L}'$ of $\mathcal{L}$ and a consistent set Δ of $\mathcal{L}'$ -formulas with the following properties:

property (i) $\Gamma \subset \Delta$

property (ii) Δ is "maximal" in the sense that either $\varphi \in \Delta$ or $\neg\varphi \in \Delta$

property (iii) For every $\mathcal{L}$ -formula φ , we have a constant symbol $c \in \mathcal{L}'$ such that $\exists x\varphi \rightarrow \varphi_c^x \in \Delta$ (note that $\exists x\varphi$ is shorthand for $\neg\forall\neg\varphi$).

The general idea of Δ is that it's a consistent set that tells you all the information needed to construct a structure $\mathfrak{A}$ in the expanded language $\mathcal{L}'$ that we can use as a structure in the original language $\mathcal{L}$. The new constant symbols c_n , we call **Henkin.Witnesses**, since essentially they allow us to "build elements" in the structure that we create that behave in the way we desire (see **Step 4**).

Step 1. Using the assumption that the set of wff's in $\mathcal{L}$ is countable, we expand the language

$\mathcal{L}$ to $\mathcal{L}' := \mathcal{L} \cup \{c_n\}_{n \in \mathbb{N}}$, where c_n for each $n \geq 1$ is a constant symbol distinct from any symbol in the original language $\mathcal{L}$. We then enumerate the well-formed $\mathcal{L}'$ -formulas $\varphi_1, \varphi_2, \dots$, and for each $n \geq 1$,

Step 2. Set $\Phi := \{\exists x \varphi_n \rightarrow \varphi_{n_{c_n}}^x : n \geq 1\}$, and show that $\Gamma \cup \Phi$ remains consistent.

Step 3. Use the prime formula trick of setting

$$\text{Sen}(\Gamma \cup \Phi) := \{\text{Sen}(\varphi) : \varphi \in \Gamma \cup \Phi\},$$

where $\text{Sen}(\varphi)$ is defined in **Section 12.2** after **Definition 12.1.2** in the [Week 12-13 Office Hour Notes](#) to mean the conversion to sentential logic by treating prime formulas as distinct sentence symbols.

Using the *Completeness-Lite Theorem*, and more specifically **Corollary 13.2.3** in the Week 12-13 Office Hour Notes, we can choose a truth assignment function $v : \{A_n\}_{n \in \mathbb{N}} \rightarrow \{T, F\}$ satisfying $\text{Sen}(\Gamma \cup \Phi)$

Define

$$\Delta := \{\mathcal{L}'\text{-formulas } \phi : \bar{v}(\text{Sen}(\phi)) = T\}.$$

Then verify that Δ is a consistent set that satisfies *properties (i)-(ii)*.

Step 4. Begin construction of a $\mathcal{L}'$ -structure $\mathfrak{A}_0$ of Δ as so:

$$|\mathfrak{A}_0| := \{t : t \text{ is a term in } \mathcal{L}'\}.$$

Define $f^{\mathfrak{A}_0} := \mathfrak{E}_f$ (the set builder function of f in the inductive set of $\mathcal{L}'$ -terms) and

$$(t_1, \dots, t_n) \in P^{\mathfrak{A}_0} \iff Pt_1 \dots t_n \in \Delta.$$

If $\mathcal{L}$ includes equality, define the additional relation E (which in the next step helps us determine equality $=$) as follows:

$$(t_1, t_2) \in E \iff t_1 t_2 \in \Delta.$$

Note that we can verify (using the Equality Axioms in Δ) that E is an **equivalence relation** (important in the next step!)

Step 5. If equality isn't included in $\mathcal{L}$, we define our final structure $\mathfrak{A}$ to be exactly the same as our previous structure $\mathfrak{A}_0$. Otherwise, define $\mathfrak{A}$ the quotient $\mathcal{L}'$ -structure $\mathfrak{A}_0 / E$ to be as so

$|\mathfrak{A}_0 / E| := \{[t] : t \text{ is an } \mathcal{L}'\text{-term}\}$, where $[t] := \{t' : (t, t') \in E\}$ is the set of all equivalence classes of E in $\mathfrak{A}$,

and define

$$f^{\mathfrak{A}}([t_1], \dots, [t_n]) = [f^{\mathfrak{A}_0}(t_1, \dots, t_n)],$$

$$([t_1], \dots, [t_n]) \in P^{\mathfrak{A}} \iff (t_1, \dots, t_n) \in P^{\mathfrak{A}_0},$$

and show that these definitions are well-defined, i.e., that for all $t_k' \in [t_k]$ ($1 \leq k \leq n$) we have the following invariance properties within the equivalence classes:

$$(a) f^{\mathfrak{A}}(t_1', \dots, t_n') \in [f^{\mathfrak{A}_0}(t_1, \dots, t_n)],$$

$$(b) (t_1', \dots, t_n') \in P^{\mathfrak{A}_0} \iff (t_1, \dots, t_n) \in P^{\mathfrak{A}_0}.$$

Step 6. Show that $\mathfrak{A}$ (as defined in **Step 5**) satisfies Δ with the variable assignment function $s : V \rightarrow |\mathfrak{A}|$ by $s = [-] \circ s_0$, where $s : V \rightarrow |\mathfrak{A}_0|$ is defined by $s_0(x) = x$ (note that we show in the first problem of Homework 13 that $\overline{s_0}(t) = t$, for every term t), which similarly shows that $s(t) = [t]$. This allows us to conclude that $\mathfrak{A}$ and s satisfy Γ in the original language $\mathcal{L}$.

Doing this in detail follows by taking $\phi \in \Delta$ and showing that $\models_{\mathfrak{A}} \phi[s]$ by structural induction on ϕ . $\square$

Note again that this proof of the completeness only holds for countable languages $\mathcal{L}$, and to generalize this proof in the setting where $\mathcal{L}$ is uncountable, we need what's called the **Well-Ordering Theorem**, which I'll briefly explain during the optional office hours.